

[DC] Exploring and Designing VR Locomotion Method based on Bio-signal for Hands-free Context and its Improvement

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ABSTRACT

Along with the advanced hand-tracking algorithms, various interactions based on hand motion are actively developing. However, if too much function are concentrated on the hand, it might be a burden on the hand and interrupt immersion. Therefore, we focused on designing a locomotion method, which is essential for exploring the broad virtual world efficiently and comfortably. We utilized bio-signal (i.e., eye tracking, EEG) to reduce the load on the hand. First, we explored to compare the usability and efficiency of our method to hand-tracking-based locomotion methods. The result showed that our methods are suitable for hands-free VR contexts. Also, we found that EEG based method proposes enhanced experience when it is used with eye-tracking. In order to improve this effect, we are currently working on research that develops user-friendly Steady-State Visual Evoked Potential (SSVEP) stimuli that suit the VR HMD format. From our research, we aim to propose design guidelines for presenting appropriate locomotion methods depending on the various contexts in the virtual environment for an enhanced hands-free VR experience.

Keywords: Virtual reality, Locomotion, Brain-Computer Interface, SSVEP, EEG, Eye-tracking, Multimodal Input

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies;

1 INTRODUCTION

Recently, an advanced head-mounted display (HMD) for both virtual and augmented reality (VR, AR) has been developed. Based on the tracking technology, these devices provide novel interaction using hands [7] and detect surroundings for spatial computing [10]. However, despite the expanded virtual world, there are still limitations in both technological and experiential approaches for users to fully explore the VR and AR environment. For instance, VR motion sickness, a critical issue in VR, interrupts users from exploring the virtual space [4] actively.

In order to overcome this issue, there are various approaches to developing novel locomotion methods that could assist users to explore virtual environments freely and comfortably [2]. Among those approaches, teleportation is the method that causes VR motion sickness the least, and thus, it is widely used in applications. Until now, teleport was utilized in targeting and triggering sequences based on controller tracking and button inputs. With the advanced tracking technology, most of the research and applications are using hand pinch and pose for teleportation targeting and triggering [3].

Hand tracking provides comfortable and accurate manipulation for UI selection, teleportation, and grabbing objects in VR [7]. However, if too many roles are assigned to the hand, it could burden users

(i.e., fatigue, complexity). For instance, one or both hands might not be available for locomotion when users are conducting other tasks with their hands. Therefore, developing and exploring hands-free interaction is essential to provide appropriate methods depending on the context.

Through our research, we propose methods that could be utilized in a hands-free context and relieve the burden of hands based on bio-signals that are achieved from the embedded sensors such as eye-tracker and electroencephalography (EEG) [1]. We utilized eye-tracking to target the location for teleportation and used the SSVEP method, the traditional brain signal detection method used in the Brain-Computer Interface (BCI) field, to trigger the teleportation. We have conducted a study comparing the usability and efficiency of these methods with hand pose-based locomotion methods. Moreover, we are currently conducting research to improve the usability of SSVEP stimuli in VR for more general usage.

2 RELATED WORKS

There have been various approaches to designing VR locomotion methods for users to explore virtual environments more comfortably [2]. These locomotion methods are classified into two categories: continuous and non-continuous ways. Continuous methods include real walking or using a joystick, which could cause VR motion sickness due to sensory mismatch. On the other hand, non-continuous methods (i.e., teleportation) have been commonly used in VR applications since these methods cause less VR sickness and are free from limited traversable space of real-world space.

The teleportation methods consist of targeting and triggering stages. Based on this process, various research is being conducted to develop more interactive and immersive teleportation methods [2, 9, 3]. For instance, a method operates teleportation with the foot movement (i.e., tilting, lifting, distance between foot) [9]. In addition, there are also attempts to utilize bio-signal data for teleportation. Li et al. utilized eye gaze for navigation while pressing a button to move forward [6]. These studies focused on proposing novel teleportation methods and comparing their method with the previous baseline method (controller-based) to verify their approach. However, to apply these methods to applications successfully, these methods should be evaluated in various environments and contexts.

Moreover, recent VR HMDs are trying to embed EEG to receive brain signals and evaluate user status or develop novel BCI interactions [1]. Among these approaches, SSVEP is a commonly used method in BCI to control a robot arm or manipulate interface (i.e., selection, typing) [8]. SSVEP is a method of detecting brain waves in the occipital lobe when a user gazes at a visual stimulus that blinks at a certain frequency. Advanced machine learning algorithms have significantly improved brain signal detection speed and accuracy. However, these methods have yet to be utilized in VR compared to 2D environments, and the current form of SSVEP stimuli is not comfortable and compatible with the HMD platform. Based on the improvement of the sensor and algorithms, BCI has strong potential to be utilized in VR, but it needs to be studied more to improve its usability for general use. Therefore, we are developing a teleportation method with SSVEP and evaluating this method in various contexts to gain insight to enhance its usability.

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3 RESEARCH PROGRESS

First, we conducted an experiment to explore the efficiency and usability of current teleportation methods. We compared three targeting methods (controller, head-direction, and eye-tracking) in a static environment [5] and a dynamic environment. Through the experiment, we found a difference in performance depending on the context and verified that the appropriate locomotion method is required to provide an immersive virtual space exploration experience.

Next, we designed bio-signal-based teleportation methods (see Fig 1) and conducted a user study to compare them with the hand-based methods. The teleportation methods were designed by combining eye-tracking, EEG (SSVEP), and hand-tracking for targeting teleport locations or triggering a teleportation event [3]. We tested these methods with the static teleportation task and assessed their elapsed time, accuracy, and usability. From the experiment result, we verified that all methods are suited for a hands-free VR context, and eye-tracking is comparatively more effective for locating a target, whereas hand-pose is comparatively more effective for initiating teleportation. In addition, interestingly, we found that eye-tracking targeting compensates for the SSVEP method limitation and improves usability. However, novel SSVEP stimuli that suit the VR HMD environment need to be developed and explored to enhance this synergistic effect between eye-tracking and EEG methods.

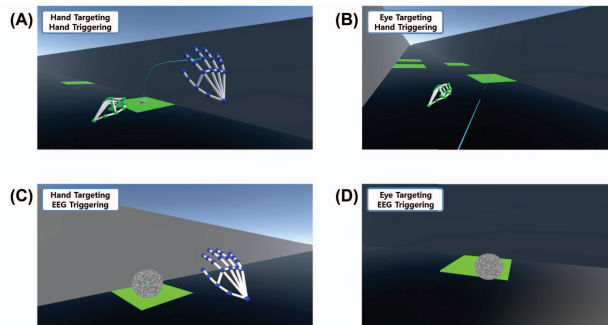


Figure 1: An illustration of the VR teleportation method (Targeting-Triggering pair). (A) Hand Movement & Hand Pose, (B) Eye-tracking & Hand Pose, (C) Hand Movement & SSVEP, and (D) Eye-tracking & SSVEP.

4 RESEARCH PLAN

Based on the insight and limitations from the exploration study, we are developing user-friendly SSVEP stimuli for VR HMD. The SSVEP stimuli are presented through the flickering mesh of the target object in the virtual world and the stimuli are mostly placed at the center of the vision to achieve a strong brain signal pattern [8]. These flickering stimuli could cause eye fatigue and distract immersion. Therefore, we are designing SSVEP stimuli that utilize peripheral areas in HMD, an area that has been not used and pointed out as a limitation of VR HMD. LED was placed around the lens to present flickering stimuli in the peripheral area, and it is activated when users gaze at the target object.

Before conducting a user study, we collected brain signals using our method (peripheral) and previous SSVEP stimuli (center) to evaluate detection performance offline. After analyzing the performance, we plan to conduct an online user study to evaluate our method in various VR contexts, such as static or dynamic moving target selection. Through the user study, we will verify that our approach has successfully provided less distracting SSVEP stimuli and an immersive experience compared to the previous method.

5 DISCUSSION

Through this study, we explored a teleportation method based on bio-signal (i.e., eye-tracking, EEG) to ease the load on the hand-tracking-based interaction. First, we designed a VR teleportation method based on eye gaze targeting and SSVEP triggering and compared it with the previous hand-based teleportation method. We found that the usability and efficiency of each method differ depending on the distance. Also, most interestingly, we found that SSVEP shows better performance when used with eye gaze targeting than hand targeting. Based on the results, we designed user-friendly SSVEP stimuli compatible with the VR HMD environment to enhance the synergistic effect. Currently, we are conducting an offline SSVEP study to examine the brain signal detection performance and planning an online user study to verify our approach. Accordingly, we would appreciate any advice or comments:

- What features could be additionally evaluated during an online VR user test to verify that our approach has improved usability rather than detecting performance or questionnaire?
- We would like to conduct behavioral analysis for user modeling and design guidelines in the future. However, since SSVEP requires more time than pressing a button and focuses more on triggering an event, conducting experiments based on tasks like Fitts' Law might be inappropriate. What would be more suitable for this type of task?

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